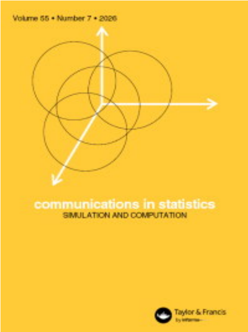

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A novel Ridge-Hurdle Poisson model for addressing multicollinearity in zero-inflated count data

HM Nayem and B. M. Golam Kibria

Department of Mathematics and Statistics, Florida International University, Miami, Florida, USA

ABSTRACT

Zero-inflated count data are prevalent in various fields, often accompanied by overdispersion and multicollinearity among predictors, posing challenges for standard modeling approaches. While models like Zero-Inflated Poisson (ZIP), Zero-Inflated Negative Binomial (ZINB), and Hurdle Poisson (HP) address excess zeros and overdispersion, they fall short under multicollinearity. This study proposes a novel Ridge-Hurdle Poisson (RHP) model that incorporates L_2 regularization into the truncated count component of the Hurdle framework to simultaneously tackle zero inflation, overdispersion, and multicollinearity. A comprehensive Monte Carlo simulation was conducted under various scenarios involving different sample sizes, predictor correlations, overdispersion levels, and structural zero probabilities. Simulation results consistently showed that the RHP model achieved the lowest mean squared error (MSE) compared to ZIP, ZINB, HP, and ridge-regularized ZIP (Zero-Inflated Poisson Ridge), especially in small samples and high multicollinearity settings. Two real-world datasets, Plastic Plywood and bioChemists, validated the model's superior predictive performance, with RHP yielding the lowest MSE in both cases. The findings highlight the robustness and practical utility of the proposed model. Future research may extend with plans to develop an R package to facilitate broader application.

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1. Introduction

Count data are prevalent across a variety of fields, including transportation safety, epidemiology, criminology, public health, ecology, and insurance. These data typically reflect the frequency of event occurrences (such as traffic accidents, disease cases, or insurance claims) and, as a result, consist of non-negative integers. Traditional approaches to count data modeling predominantly utilize the Poisson regression model, which assumes that the response variable follows a Poisson distribution and that the mean equals the variance. While this model is straightforward and interpretable, it often proves inadequate in practical situations where the variance of the response variable exceeds the mean an issue referred to as overdispersion (Lord and Mannering 2010; Cameron and Trivedi 2013).

To address overdispersion, researchers commonly utilize the Negative Binomial (NB) model, as it incorporates a dispersion parameter that accounts for variability among observations (Asar 2018). While the NB model provides greater flexibility, many datasets also exhibit another frequent characteristic: zero inflation. Zero inflation occurs when the count of observed zeros greatly exceeds what would be expected under a standard Poisson or NB distribution. This excess of zeros can be attributed to two distinct mechanisms: (i) structural zeros, which arise when an

CONTACT HM Nayem hnaye001@fiu.edu Department of Mathematics and Statistics, Florida International University, 11200 SW 8th Street, Miami, FL 33199, USA

event is inherently impossible for a specific subset of the population, and (ii) sampling zeros, which occur due to random chance (Lambert 1992; Greene 1994; Ridout et al. 1998). A Hurdle model is also widely recognized as an effective approach for handling both underdispersion and overdispersion in count data (Altinisik 2023)

To effectively model zero inflation, two-component models have been developed. The Zero-Inflated Poisson (ZIP) and Zero-Inflated Negative Binomial (ZINB) models suggest that the data is derived from a combination of two processes: a binary process that identifies whether an observation is categorized as a structural zero, and a count process (Poisson or NB) that generates counts, including potential zeros. These models have been widely applied in various fields, including highway safety (Lord et al. 2005; Karim et al. 2020), health science (Deb and Trivedi 1997; Abonazel et al. 2021), and ecological species modeling (Martin et al. 2005), where the occurrence of excess zeros is prevalent. Unlike ZIP/ZINB models that allow for zeros from both components, the Hurdle model assumes that all zeros arise from the binary component. The count component then focuses solely on modeling positive counts using a truncated Poisson (or NB) distribution (Mullahy 1986; Washington et al. 2020; Lee et al. 2021).

In addition to issues of overdispersion and zero inflation, multicollinearity among predictor variables presents another significant challenge in count regression analysis. Multicollinearity arises from high correlations among independent variables, which can inflate the standard errors of regression coefficients, resulting in inference that is both unstable and unreliable. This problem is particularly pronounced in high-dimensional datasets or observational studies, where controlling for covariate selection is often difficult (Montgomery et al. 2012; Dormann et al. 2013). Traditional methods such as variable selection or principal component regression frequently compromise interpretability or risk omitting important covariates.

To address the challenges of multicollinearity and enhance predictive performance, regularization techniques have emerged as crucial tools. Ridge regression (Hoerl and Kennard 1970) incorporates an L_2 penalty that shrinks coefficients toward zero without removing them, thereby stabilizing estimates in the presence of correlated predictors. In the domain of count data, recent studies have adapted regularization methods for generalized linear models (GLMs), such as penalized Poisson and NB regression (Kibria et al. 2015; Rady et al. 2018; Khan et al. 2023). For instance, Park and Hastie (2007) developed path algorithms for penalized GLMs, while Wang, Wang, et al. (2020) reviewed the applications of regularization in count regression. Penalized models have shown promising improvements in predictive accuracy, model interpretability, and variable stability in high-dimensional or multicollinear scenarios (Viallon and Commenges 2012; Kharrat et al. 2022). These advancements have also been extended to zero-inflated models, where penalization has been applied to both the zero and count components (Ye and Pan 2006; Friedman et al. 2010; Almulhim et al. 2025).

Despite the significant advancements made in the field, Hurdle models have received relatively little attention in the penalized regression literature. While several studies have introduced regularized ZIP and ZINB models (Kibria et al. 2011, 2013; Yüzbaşi and Asar 2018; Qasim et al. 2020; Aladeitan et al. 2021; Kim et al. 2021; Kwon and Hwang 2022; Akram et al., 2024; Akram et al. 2025), there has been limited exploration of penalized Hurdle models, particularly in the context of both zero inflation and multicollinearity. Moreover, to our knowledge, no existing research has proposed or systematically evaluated a Ridge-regularized Hurdle Poisson (HP) model, which represents a notable gap in the toolkit available for analyzing complex count data.

This study is driven by the necessity to address a significant methodological gap in the literature: we propose an innovative Ridge-regularized HP model that incorporates L_2 regularization into the truncated Poisson count component of the Hurdle framework. The primary objective of this research is to systematically compare the performance of several count data regression models, specifically the ZIP, ZINB, HP, Ridge-regularized ZIP, and the proposed Ridge-regularized HP model under various data conditions. Utilizing a comprehensive simulation framework, we evaluate the performance using mean squared error (MSE) of these models across a range of scenarios defined by different levels of zero inflation, overdispersion, and multicollinearity among predictors, and sample sizes.

The structure of the article is as follows. Section 2 discusses the methodological framework and introduces the proposed Ridge-Hurdle Poisson (RHP) model. Section 3 details the Monte Carlo simulation design and presents the corresponding results. Section 4 offers real-data validation through two empirical applications, and Sec. 5 concludes with the main findings and avenues for future research.

2. Methodology

To accomplish the objectives of this study, we conducted a systematic comparison of the performance of various count data regression models, including the ZIP, ZINB, HP, Ridge-regularized ZIP, and the newly proposed Ridge-regularized HP model, under diverse data conditions. This section outlines the methodological framework in three key parts: first, we present the theoretical foundations of both standard and regularized count models; second, we describe a comprehensive Monte Carlo simulation designed to evaluate model performance under scenarios characterized by excess zeros and multicollinearity; and finally, we validate the simulation findings using a real-world dataset. All analyses were performed using the R programming language to ensure reproducibility and flexibility in model implementation.

2.1. Zero-Inflated Poisson (ZIP)

Let $Y_i \in \mathbb{N}_0$ denote a count response variable exhibiting excess zeros. The ZIP distribution (Lambert 1992) posits that every observation is sampled from a combination of a degenerate distribution at zero (with probability π_i) and a Poisson distribution (with probability $1-\pi_i$). The probability mass function (PMF) of the Poisson distribution with mean $\lambda_i > 0$ is

$$P(Y_i = y_i) = \frac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!}, y_i \in \mathbb{N}_0$$

In the ZIP model, $\lambda_i = \exp\left(x_i^\top \beta\right)$, and $\pi_i = \frac{\exp\left(z_i^\top \gamma\right)}{1+\exp\left(z_i^\top \gamma\right)}$.
Then, the PMF becomes

$$P(Y_i = y_i) = \begin{cases} \pi_i + (1-\pi_i)e^{-\lambda_i}, & \text{if } y_i = 0 \\ (1-\pi_i)\dfrac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!}, & \text{if } y_i > 0 \end{cases}$$

Let $X \in \mathbb{R}^{n\times p}$ and $Z \in \mathbb{R}^{n\times q}$ be the covariate matrix for the count and zero components, respectively. Let $\theta = (\beta, \gamma)$. The log-likelihood for independent observations $\{y_i\}_{i=1}^n$ is

$$\log\ L(\theta) = \sum_{i:y_i=0} \log\left[\pi_i + (1-\pi_i)e^{-\lambda_i}\right] + \sum_{i:y_i>0} \left[\log(1-\pi_i) - \lambda_i + y_i \log\lambda_i - \log y_i\right]$$

Let W_λ be the diagonal matrix of Poisson weights:

$$W_\lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$$

Let the score vector and Fisher information matrix for the Poisson part (conditional on non-zero counts) be approximated as

$$U_\beta = X^\top(y-\lambda), I_\beta = X^\top W_\lambda X$$

Then the approximate estimator:

$$\hat{\beta} = \left(X^\top W_\lambda X\right)^{-1} X^\top W_\lambda y$$

Then the MSE matrix is

$$\mathrm{MSE}\left(\hat{\beta}\right) = \mathbb{E}\left[\left(\hat{\beta} - \beta\right)\left(\hat{\beta} - \beta\right)^\top\right] = \mathrm{Var}\left(\hat{\beta}\right) + \mathrm{Bias}\left(\hat{\beta}\right)\mathrm{Bias}\left(\hat{\beta}\right)^\top$$

If $\hat{\beta}$ is asymptotically unbiased, then

$$\mathrm{MSE}\left(\hat{\beta}\right) \approx I_\beta^{-1} = \left(X^\top W_\lambda X\right)^{-1}$$

2.2. Zero Inflated-Negative Binomial (ZINB)

Let $Y_i \in \mathbb{N}_0$ be a count response exhibiting overdispersion and excess zeros. The ZINB model (Lambert 1992; Raihan et al. 2019) extends the ZIP model by using the NB distribution for the count component to accommodate overdispersion.

Let $Y_i \sim \mathrm{NB}(\mu_i, \phi)$, where $\mu_i = \mathbb{E}[Y_i]$ and ϕ^{-1} is the dispersion parameter.

Then, the NB PMF is (Akram et al. 2022)

$$P(Y_i = y_i) = \frac{\Gamma(y_i + \phi)}{\Gamma(y_i + 1)\Gamma(\phi)} \left(\frac{\mu_i}{\mu_i + \phi}\right)^{y_i} \left(\frac{\phi}{\mu_i + \phi}\right)^{\phi}, y_i \in \mathbb{N}_0$$

In the ZINB model, the response variable Y_i equals zero with structural probability p_i, while with probability $1 - p_i$, Y_i follows a Negative Binomial distribution with mean μ_i and dispersion parameter ϕ. As a result, the complete PMF can be represented as a finite combination of a point mass at zero along with a standard NB distribution:

$$P(Y_i = y_i) = \begin{cases} p_i + (1 - p_i)\left(\dfrac{\phi}{\mu_i + \phi}\right)^{\phi}, & \text{if } y_i = 0 \\ (1 - p_i) \cdot \dfrac{\Gamma(y_i + \phi)}{\Gamma(y_i + 1)\Gamma(\phi)} \left(\dfrac{\mu_i}{\mu_i + \phi}\right)^{y_i} \left(\dfrac{\phi}{\mu_i + \phi}\right)^{\phi}, & \text{if } y_i > 0 \end{cases}$$

where $\mu_i = \exp\left(x_i^\top \beta\right)$ and $p_i = \frac{\exp\left(z_i^\top \gamma\right)}{1+\exp\left(z_i^\top \gamma\right)}$.

Let $\theta = \left(\phi, \beta^\top, \gamma^\top\right)^\top \in \mathbb{R}^{1+p+q}$. Given data $\{y_i, x_i, z_i\}_{i=1}^n$, the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \mathbb{I}(y_i = 0)\log\left[p_i + (1 - p_i)\left(\frac{\phi}{\mu_i + \phi}\right)^{\phi}\right] + \sum_{i=1}^{n} \mathbb{I}(y_i > 0)\left[\log(1 - p_i) + \log\left(\frac{\Gamma(y_i + \phi)}{\Gamma(y_i + 1)\Gamma(\phi)}\right) + y_i\log\left(\frac{\mu_i}{\mu_i + \phi}\right) + \phi\log\left(\frac{\phi}{\mu_i + \phi}\right)\right]$$

The observed information (negative Hessian) is

$$I_\beta = X^\top W X$$

where W contains the second derivatives of the likelihood w.r.t. β.

$$\hat{\beta} \approx \left(X^\top W X\right)^{-1} X^\top W \tilde{y}$$

where $\tilde{y}$ is a pseudo-response vector.

Thus, the asymptotic MSE of $\hat{\beta}$ is

$$\text{MSE}\left(\hat{\beta}\right) = \mathbb{E}\left[\left(\hat{\beta} - \beta\right)\left(\hat{\beta} - \beta\right)^{\top}\right] = \text{I}_{\beta\beta}^{-1}$$

where $I_{\beta\beta}$ is the $p \times p$ block of the full Fisher information matrix $I(\theta)$ defined as

$$I(\theta) = -\frac{\partial^2 \ell(\theta)}{\partial\theta\partial\theta^{\top}} = \begin{bmatrix} L_{\phi\phi} & L_{\phi\beta} & L_{\phi\gamma} \\ \cdot & L_{\beta\beta} & L_{\beta\gamma} \\ \cdot & \cdot & L_{\gamma\gamma} \end{bmatrix}$$

2.3. Hurdle Poisson (HP)

Hurdle models provide an alternative to zero-inflated models for count data with an excess of zeros. Unlike zero-inflated models, which treat zeros as arising from both structural and sampling processes, hurdle models consider zero and positive counts as resulting from two separate mechanisms. A binary component models the hurdle (zero), while a truncated count model addresses the positive outcomes. Introduced by Mullahy in 1986 and refined by Cameron and Trivedi in 1998, this framework allows the hurdle to be crossed with a certain probability, after which positive counts are generated from a zero-truncated count distribution (Mullahy 1986; Cameron and Trivedi 1998).

Let $f_1(y)$ denote the binary (hurdle) process and $f_2(y)$ denote the count model. Then, the Hurdle PMF is defined as (Rose et al. 2006)

$$P(Y = y) = \begin{cases} f_1(0) = p, & y = 0 \\ (1-p) \cdot \dfrac{f_2(y)}{1 - f_2(0)} = (1-p) f_2^*(y), & y > 0 \end{cases}$$

where $f_2^*(y)$ is the zero-truncated version of the count model $f_2(y)$ and $p = \mathbb{P}(Y = 0)$ from the hurdle (binary) process.

This formulation ensures that all zeros are governed by the binary process, and positive counts arise only from the truncated distribution. If the count component $f_2(y)$ is Poisson distributed with mean μ, and the hurdle is modeled using a Bernoulli-logistic function $p_i = \Pr(Y_i = 0)$. Then, the HP model PMF becomes

$$P(Y_i = y_i) = \begin{cases} p_i, & y_i = 0 \\ (1-p_i) \cdot \dfrac{e^{-\mu_i}\mu_i^{y_i}}{(1 - e^{-\mu_i}) y_i!}, & y_i > 0 \end{cases}$$

This is a two-part model:

1. Binary model: $\log\left(p_i/(1-p_i)\right) = z_i^{\top}\gamma$

2. Truncated Poisson model: $\mu_i = \exp\left(x_i^{\top}\beta\right)$ where $x_i \in \mathbb{R}^p$ are the covariates for the count component and $z_i \in \mathbb{R}^q$ are the covariates for the binary hurdle component.

Given independent observations (y_i, x_i, z_i), the log-likelihood is

$$\ell(\theta) = \sum_{i=1}^{n} \mathbb{I}(y_i = 0)\log(p_i) + \sum_{i=1}^{n} \mathbb{I}(y_i > 0)\left[\log(1-p_i) + \log\left(\frac{e^{-\mu_i}\mu_i^{y_i}}{(1 - e^{-\mu_i}) y_i}\right)\right]$$

Substituting the logistic and Poisson specifications gives

$$\ell(\theta) = \sum_{i=1}^{n} \mathbb{I}(y_i = 0)\left[-\log\left(1 + e^{z_i^\top \gamma}\right)\right]$$
$$+ \sum_{i=1}^{n} \mathbb{I}(y_i > 0)\left[-\log\left(1 + e^{z_i^\top \gamma}\right) + x_i^\top \beta y_i - e^{x_i^\top \beta} - \log\left(1 - e^{-e^{x_i^\top \beta}}\right) - \log(y_i)\right]$$

Letting $\theta = \left(\beta^\top, \gamma^\top\right)^\top$, this function is maximized numerically to estimate $\hat{\theta}$.
The expected value of the HP is

$$\mathbb{E}[Y_i] = \frac{(1 - p_i)\mu_i}{1 - e^{-\mu_i}}$$

The variance can be derived using the law of total variance:

$$\mathrm{Var}(Y_i) = (1 - p_i)\left[\frac{\mu_i^2 + \mu_i}{1 - e^{-\mu_i}} - \left(\frac{\mu_i}{1 - e^{-\mu_i}}\right)^2\right] + p_i \cdot \left(\frac{\mu_i}{1 - e^{-\mu_i}}\right)^2$$

MSE of the RHP estimator, eigen basis:

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{RHP}}\right) = \tau \sum_{j=1}^{p} \frac{\lambda_j^{(\mathrm{RHP})}}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2} + k^2 \sum_{j=1}^{p} \frac{\alpha_j^2}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2}$$

2. 4. Regularization methods

Let $Y \in \mathbb{R}^n$ be a vector of count responses (non-negative integers), $X \in \mathbb{R}^{n \times p}$ be the design matrix of predictors, and $\beta \in \mathbb{R}^p$ be the coefficient vector. For count data, a common modeling framework is the Poisson regression model, which assumes that each observation $y_i \sim \mathrm{Poisson}(\mu_i)$, where the mean μ_i is related to the predictors through a log link function:

$$\log(\mu_i) = x_i^T \beta \text{ or equivalently } \mu_i = \exp\left(x_i^T \beta\right)$$

$$\ell(\beta) = \sum_{i=1}^{n} \left(y_i x_i^T \beta - \exp\left(x_i^T \beta\right) - \log(y_i!)\right)$$

However, when the predictors in X are highly correlated (i.e. multicollinearity), or when $p > n$, the MLE can be unstable and lead to overfitting (Park and Hastie 2007; Friedman et al. 2010). To tackle this, regularization methods like Ridge, Lasso, and Elastic Net impose penalties on coefficient sizes to improve estimation stability and prediction accuracy.

2.4.1. Ridge regression

Hoerl and Kennard (1970) were the pioneers of the ridge regression estimator, addressing the problem of multicollinearity. This method aims to minimize the total of squared residuals while placing a restriction on the overall sum of the squares of the coefficients (Hoerl and Kennard 1970). While their formulation focused on continuous responses, the ridge regression concept can be extended to count data by incorporating the regularization framework into GLMs, particularly the Poisson regression model. When the response variable is a count, the standard modeling approach is Poisson regression, which assumes that the response Y_i follows a Poisson distribution with mean $\mu_i = \exp\left(x_i^T \beta\right)$. However, in the presence of multicollinearity among the predictors, the MLEs of the coefficients β can become unstable and highly variable. Ridge regression addresses this issue by introducing a penalty term that shrinks coefficient estimates toward zero, reducing variance while introducing some bias.

In the context of count data, the ridge-penalized Poisson regression minimizes the penalized negative loglikelihood:

$$\hat{\beta}_{\text{ridge}} = \underset{\beta}{\operatorname{argmin}}\left\{-\ell(\beta) + \lambda\|\beta\|_2^2\right\}$$

where $\ell(\beta)$ is the log-likelihood of the Poisson model:

$$\ell(\beta) = \sum_{i=1}^{n}\left(y_i x_i^T\beta - \exp\left(x_i^T\beta\right) - \log(y_i!)\right)$$

$\|\beta\|_2^2 = \sum_{j=1}^{p}\beta_j^2$ is the squared L_2-norm penalty and $\lambda \geq 0$ is the ridge (regularization) parameter that controls the strength of the shrinkage.

When $\lambda = 0$, the estimator reduces to the standard MLE for Poisson regression. As λ increases, the coefficients are increasingly shrunk toward zero.

Ridge regression mitigates multicollinearity by constraining coefficient estimates, which can reduce the estimator's variance while introducing some bias (Akram et al., 2024).

2.4.2. Zero-Inflated Poisson Ridge

The Zero-Inflated Poisson Ridge (ZIPR) model extends the standard ZIP model by addressing multicollinearity through the inclusion of an ℓ_2 penalty term (ridge). The ZIP model is appropriate for count data exhibiting excessive zeros and assumes that for each observation y_i, the distribution is given by Al-Taweel and Algamal (2022):

$$P(Y_i = y_i) = \begin{cases} \pi_i + (1-\pi_i)e^{-\mu_i}, \ y_i = 0 \\ (1-\pi_i)\dfrac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}, \ y_i > 0 \end{cases}$$

where $\pi_i = \exp\left(z_i^\top\gamma\right)/\left(1+\exp\left(z_i^\top\gamma\right)\right), \mu_i = \exp\left(x_i^\top\beta\right)$, and x_i, z_i are covariate vectors for the count and zero components, respectively.The ridge estimator is obtained by maximizing a penalized log-likelihood function:

$$\ell_{\text{pen}}(\beta,\gamma) = \ell(\beta,\gamma) - \lambda\beta^\top\beta$$

where $\ell(\cdot)$ is the ZIP log-likelihood and $\lambda > 0$ is the ridge penalty parameter. The ridge estimator for β under the ZIPR model is given by

$$\hat{\beta}_{\text{ZIPR}} = \left(X^\top WX + \lambda I_p\right)^{-1}X^\top WY^*$$

where $X \in \mathbb{R}^{n\times p}$ is the design matrix for the count component, $W \in \mathbb{R}^{n\times n}$ is the weight matrix derived from the second derivatives of the log-likelihood (Fisher scoring or IRLS), $Y^* \in \mathbb{R}^n$ is the adjusted response vector, $\lambda \in \mathbb{R}_+$ is the tuning (ridge) parameter, and I_p is the identity matrix of dimension $p \times p$.

Under standard regularity conditions and assuming asymptotic normality of the ridge estimator, the asymptotic MSE of $\hat{\beta}_{\text{ZIPR}}$ is given by

$$\text{MSE}\left(\hat{\beta}_{\text{ZIPR}}\right) = \left(X^\top WX + \lambda I_p\right)^{-1}X^\top W\Sigma WX\left(X^\top WX + \lambda I_p\right)^{-1}$$

where $\Sigma = \text{Var}(Y^*)$ is the working variance of the transformed response.When $W = I_n$, this simplifies to the standard ridge MSE form:

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{ZIPR}}\right) = \sigma^2\left(X^\top X + \lambda I_p\right)^{-1} X^\top X\left(X^\top X + \lambda I_p\right)^{-1}$$

Then, in the eigen basis, the MSE becomes

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{ZIPR}}\right) = \sigma^2\sum_{j=1}^{p}\left[\frac{\lambda_j}{(\lambda_j+\lambda)^2}\right] + \lambda^2\sum_{j=1}^{p}\frac{\alpha_j^2}{(\lambda_j+\lambda)^2}$$

2.4.3. Proposed Ridge-Hurdle Poisson (RHP)

Let $\hat{\beta}_{\mathrm{HP}}$ denote the MLE obtained from the positive-count component of the HP model. The proposed RHP estimator introduces a ridge penalty to control for multicollinearity or overfitting in the count component.

The following is the estimator of the RHP adding weight matrix.

$$\hat{\beta}_{\mathrm{RHP}} = \left(X^\top \hat{W} X + kI_p\right)^{-1} X^\top \hat{W} X \hat{\beta}_{\mathrm{HP}}$$

where $X \in \mathbb{R}^{n\times p}$ is the design matrix for the count component (excluding zero counts), $\hat{W} = \mathrm{diag}(\hat{\mu}_1, \ldots, \hat{\mu}_n)$ is the diagonal weight matrix with elements $\hat{\mu}_i = \exp\left(x_i^\top \hat{\beta}_{\mathrm{HP}}\right)$, $k \geq 0$ is the ridge penalty parameter, I_p is the $p \times p$ identity matrix, and $\hat{\beta}_{\mathrm{RHP}} = \hat{\beta}_{\mathrm{HP}}$ when $k = 0$.

Let the eigen-decomposition of $X^\top \hat{W} X$ be

$$X^\top \hat{W} X = \sum_{j=1}^{p} \lambda_j \psi_j \psi_j^\top$$

Let the true coefficient vector β be expressed in terms of this eigen basis:

$$\beta = \sum_{j=1}^{p} \alpha_j \psi_j, \text{ where } \alpha_j = \psi_j^\top \beta$$

Then the MSE of $\hat{\beta}_{\mathrm{RHP}}$ is

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{RHP}}\right) = \hat{\tau}\sum_{j=1}^{p}\frac{\lambda_j}{(\lambda_j+k)^2} + k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{(\lambda_j+k)^2}$$

where λ_j is the jth eigenvalue of $X^\top \hat{W} X$, ψ_j is the corresponding orthonormal eigenvector, $\alpha_j = \psi_j^\top \beta$ is the projection of the true coefficient vector onto the eigenvector ψ_j, and $\hat{\tau} = (1/(n-p-1))\sum_{i=1}^{n}(y_i - \hat{\mu}_i)^2$ is the estimated residual dispersion.

Lemma: Consider two estimators of β, denoted as $\hat{\beta}_1$ and $\hat{\beta}_2$. The estimator $\hat{\beta}_2$ is considered more efficient than $\hat{\beta}_1$ in terms of MSE if and only if $\mathrm{MSE}\left(\hat{\beta}_1\right) - \mathrm{MSE}\left(\hat{\beta}_2\right) \geq 0$ (Alheety et al. 2025).

Theorem 2.1: When $0 \leq k \leq 1$, the RHP estimator outperforms the HP estimator with respect to the MSE criterion, meaning that $\mathrm{MSE}\left(\hat{\beta}_{\mathrm{HP}}\right) - \mathrm{MSE}\left(\hat{\beta}_{\mathrm{RHP}}\right) \geq 0$.

Proof: MSE of HP estimator:

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{HP}}\right) = \hat{\tau}\sum_{j=1}^{p}\frac{1}{\lambda_j}$$

MSE of RHP estimator:

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{RHP}}\right) = \hat{\tau}\sum_{j=1}^{p}\frac{\lambda_j}{\left(\lambda_j + k\right)^2} + k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{\left(\lambda_j + k\right)^2}$$

Note that

$$\frac{\lambda_j}{\left(\lambda_j + k\right)^2} < \frac{1}{\lambda_j} \quad \text{for } \lambda_j > 0, k > 0$$

Also,

$$k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{\left(\lambda_j + k\right)^2} \geq 0$$

So

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{RHP}}\right) < \mathrm{MSE}\left(\hat{\beta}_{\mathrm{HP}}\right)$$

Thus, adding shrinkage reduces MSE when $0 < k \leq 1$. When $k = 0$, RHP reduces to HP, so the difference is 0. Therefore, the proof is complete.

Theorem 2.2: When $0 \leq k \leq 1$, the RHP estimator outperforms the ZIPR estimator with respect to the MSE criterion, meaning that MSE $\left(\hat{\beta}_{\mathrm{ZIPR}}\right)$ − MSE $\left(\hat{\beta}_{\mathrm{RHP}}\right) \geq 0$.

Proof: The MSE of the ZIPR estimator is

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{ZIPR}}\right) = \tau\sum_{j=1}^{p}\frac{\lambda_j}{\left(\lambda_j + k\right)^2} + k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{\left(\lambda_j + k\right)^2}$$

The MSE of the RHP estimator is

$$\mathrm{MSE}\left(\hat{\beta}_{\mathrm{RHP}}\right) = \tau\sum_{j=1}^{p}\frac{\lambda_j^{(\mathrm{RHP})}}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2} + k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2}$$

Assumptions for fair comparison:
1. Let both models have the same penalty term $k \geq 0$.
2. Assume the same eigen basis and the same dispersion factor τ.
3. Let $\lambda_j^{(\mathrm{RHP})} > \lambda_j$ for all j, since hurdle models use only positive counts, leading to stronger signal and better-conditioned design matrix.

First term:

$$\frac{\lambda_j^{(\mathrm{RHP})}}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2} < \frac{\lambda_j}{\left(\lambda_j + k\right)^2} \quad \text{for } \lambda_j^{(\mathrm{RHP})} > \lambda_j > 0, k > 0$$

The function

$$f(\lambda) = \frac{\lambda}{(\lambda + k)^2}$$

is strictly decreasing in λ for $\lambda > 0, k > 0$, since:

$$f'(\lambda) = \frac{(\lambda + k)^2 - 2\lambda(\lambda + k)}{(\lambda + k)^4} < 0 \text{ for all } \lambda, k > 0$$

So,

$$\frac{\lambda_j^{(\mathrm{RHP})}}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2} < \frac{\lambda_j}{(\lambda_j + k)^2} \forall j$$

Second term:
Since $\lambda_j^{(\mathrm{RHP})} > \lambda_j$, and denominators are larger:

$$\frac{\alpha_j^2}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2} < \frac{\alpha_j^2}{(\lambda_j + k)^2} \Rightarrow k^2 \sum_{j=1}^{p} \frac{\alpha_j^2}{\left(\lambda_j^{(\mathrm{RHP})} + k\right)^2} < k^2 \sum_{j=1}^{p} \frac{\alpha_j^2}{(\lambda_j + k)^2}$$

Thus, $\mathrm{MSE}(\hat{\boldsymbol{\beta}}_{\mathrm{ZIPR}}) < \mathrm{MSE}(\hat{\boldsymbol{\beta}}_{\mathrm{HP}})$ for all $0 \le k \le 1$. Therefore, the proof is completed.

3. Monte Carlo simulation

3.1. Simulation design

The process utilized for data generation in the models was executed in accordance with a widely recognized methodology, as detailed below (Kibria 2003; Hoque and Kibria 2023, 2024; Nayem et al. 2024; Yasmin and Kibria 2025).

$$x_{ij} = \sqrt{(1 - \rho^2)}\ a_{ij} + \rho a_{i,p+1}, i = 1, 2, \ldots, m; j = 1, 2, \ldots, p$$

where ρ signifies the degree of correlation coefficient between any two predictors, a_{ij} denotes the independently generated pseudo-random variable obtained from a standard normal distribution, and p represents the number of predictors.

The response variable Y_i, representing a zero-inflated count outcome, was generated through the following two-part process:

$$Y_i = \begin{cases} 0, & \text{with probability } \pi_i \\ \mathrm{NB}(\mu_i, \theta), & \text{with probability } 1 - \pi_i \end{cases}, i = 1, 2, \ldots, m$$

where π_i is the zero-inflation probability obtained from a logistic function with intercept γ_0 and $\mathrm{NB}(\mu_i, \theta)$ denotes an NB distribution with mean μ_i and overdispersion parameter θ. Specifically, the linear predictor for the count component is defined as

$$\eta_i = \beta_0 + \beta_1 Z_{i1} + \beta_2 Z_{i2} + \cdots + \beta_p Z_{ip}, i = 1, 2, \ldots, m$$

The zero-inflation probability is given by

$$\pi_i = \frac{\exp(\gamma_0)}{1 + \exp(\gamma_0)}$$

where $\gamma_0 \in \{0, 1\}$ is the intercept in the logistic model determining the structural zero probability, and $\theta \in \{1, 5\}$ is the dispersion parameter controlling the extra-Poisson variation. The count part of the model (when not structurally zero), therefore, satisfies

$$Y_i \sim \text{Negative Binomial } (\mu_i, \theta)$$

For the simulation study, a comprehensive range of data conditions was employed to assess the performance of the models under multicollinearity and overdispersion. Specifically, two values of predictors: $p = 10$ and $p = 20$, correlation coefficient ρ among explanatory variables across four levels: $0.80, 0.90$, 0.95, and 0.99; sample sizes: $n = 100, 200$, and 500. To model the excess zeros or the zero-hurdle mechanism, two intercept values in the logistic model were used: 0 and 1, where a logit intercept of 0 corresponds to a 50% probability of structural zeros, and a logit intercept of 1 corresponds to approximately 73% structural zeros (Fletcher et al. 2005; Hua et al. 2014). Overdispersion in the count component was incorporated using two values of the overdispersion parameter $\theta = 1$ and $\theta = 5$, where a higher θ indicates greater variance in the count distribution, particularly in the Negative Binomial and Hurdle models (Bertoli et al. 2020). Each simulation scenario was independently repeated $N = 5000$ times to ensure stable and reliable performance evaluation across all model configurations (Meketon 1987).

MSE is minimized when the coefficient vector β is aligned with the normalized eigenvector corresponding to the largest eigenvalue of the $X'X$ matrix. The simulation study calculates MSE by evaluating the difference between the estimated coefficient vectors β, obtained from different models, and the true β (Newhouse and Oman 1971). For each simulation, the true β is determined by identifying and normalizing the eigenvector associated with the highest eigenvalue of $X'X$, such that $\beta'\beta = 1$. The average MSE is then calculated by applying the formula consistently across all simulation runs (Nayem et al. 2025).

$$\mathrm{MSE}\left(\hat{\beta}_i^*\right) = \frac{1}{N}\sum\nolimits_{i=1}^{n}\left(\hat{\beta}_i - \beta\right)'\left(\hat{\beta}_i - \beta\right)$$

where $\hat{\beta}_i^*$ is any estimator, β is the true parameter, and $N = 5000$. Lower values of MSE indicate a better model based on MSE.

3.2. Simulation results

This section summarizes findings from a simulation study evaluating various count data models under different conditions, aiming to identify a suitable framework for zero-inflated and overdispersed count outcomes in roadway safety analysis. It explores how model performance, measured by MSE, is influenced by factors like sample size (n), number of predictors (p), predictor correlation, intercept logit, and overdispersion level, including traditional models like ZIP and ZINB as well as regularized approaches.

3.2.1. Effect of sample size on performance

An increase in sample size generally led to improved model performance across all models, as reflected by reduced MSEs. However, the RHP model consistently outperformed others, particularly in small sample conditions, demonstrating its stability and resistance to overfitting. When $p = 10$, correlation $= 0.80$, intercept logit $= 0$, and overdispersion $= 1$, the MSE of RHP was 0.0729, 0.0594, and 0.0787 for $n = 100$, 200, and 500, respectively (Table 2). In comparison, the corresponding MSEs for ZIP and ZINB were substantially higher, especially at smaller sample sizes, highlighting the robustness of our proposed model. This trend persisted across increasing correlation levels (Tables 1–4) and when $p = 20$ (Tables 5–8), where RHP maintained the lowest MSEs across varying sample sizes.

3.2.2. Effect of number of predictors on performance

Model performance generally degraded with an increase in the number of predictors. However, the RHP model was more resilient to this effect compared to others. When comparing Tables 2 and 6 (both with correlation $= 0.80$ and intercept logit $= 0$), the MSE for RHP increased modestly from 0.0729 (Table 2) to 0.2934 (Table 5) for $n = 100$. In contrast, the ZIP model's MSE increased more drastically, from 0.5547 to 1.6184 under the same conditions, indicating that RHP scales better with high-dimensional data. At extreme settings (Table 8: $p = 20$, correlation $= 0.99$,

Table 1. Simulated MSE values when $p = 10$ and correlation $= 0.80$.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	0.5547	0.2721	0.2018	0.2684	0.1189	0.1060
ZINB		0.4486	0.1629	0.1106	0.2677	0.0998	0.0864
HP		0.5733	0.2468	0.1994	0.2146	0.1134	0.1069
RHP		0.0729	0.0594	0.0787	0.0667	0.0547	0.0545
ZIPR		0.1221	0.0999	0.1148	0.1227	0.1015	0.0917
ZIP	1	4.0755	0.7030	0.2489	41.8425	0.1890	0.1230
ZINB		4.5034	0.5299	0.1314	39.9644	0.1760	0.1017
HP		3.2034	0.5229	0.2308	5.7094	0.1786	0.1174
RHP		0.0492	0.0410	0.0578	0.0286	0.0429	0.0422
ZIPR		0.0916	0.1218	0.1665	0.1367	0.1088	0.1110

Table 2. Simulated MSE values when $p = 10$ and correlation $= 0.90$.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	0.8023	0.4760	0.3734	0.3344	0.1537	0.1255
ZINB		0.7117	0.2033	0.1360	0.3202	0.1138	0.0938
HP		0.9052	0.4399	0.3798	0.2931	0.1583	0.1274
RHP		0.2520	0.0680	0.1314	0.1217	0.0799	0.0687
ZIPR		0.4135	0.1036	0.1318	0.4688	0.0849	0.1499
ZIP	1	52.6066	1.1670	0.5445	21.2201	0.2127	0.1988
ZINB		68.9659	0.6180	0.2361	56.6104	0.2059	0.1174
HP		3.8460	1.0676	0.4729	0.8940	0.2216	0.1885
RHP		0.0189	0.0827	0.1513	0.0456	0.0597	0.0736
ZIPR		0.1072	0.1264	0.2783	0.0831	0.1009	0.1740

Table 3. Simulated MSE values when $p = 10$ and correlation $= 0.95$.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	2.3792	0.7441	0.7529	0.6768	0.2338	0.2046
ZINB		1.9245	0.4014	0.2028	0.6357	0.1546	0.1027
HP		2.3286	0.7961	0.7478	0.6842	0.2455	0.2041
RHP		0.5480	0.1664	0.3396	0.1374	0.1558	0.1223
ZIPR		0.8769	0.0914	0.3868	0.4161	0.3260	0.2605
ZIP	1	21.9415	1.7189	0.9600	31.6253	0.3317	0.2443
ZINB		21.4170	0.9025	0.3358	29.8227	0.3368	0.1430
HP		11.8809	1.8173	0.9235	1.9304	0.3676	0.2449
RHP		1.4299	0.4945	0.2784	0.1185	0.0618	0.0973
ZIPR		0.3571	0.4589	0.3280	0.0838	0.0978	0.2485

$n = 100$), RHP still yielded a considerably lower MSE (7.3989) compared to ZINB (96.5951) and HP (97.6672), underscoring its advantage in high-dimensional, highly correlated settings.

3.2.3. Effect of correlation coefficients on performance

Model performance deteriorated with increasing correlation among predictors, which typically challenges model estimation. Nevertheless, RHP maintained a substantial edge in predictive accuracy. In Table 2 (correlation = 0.80), RHP reported MSEs of 0.0729, 0.0594, and 0.0787 for $n = 100$, 200, and 500, respectively. These values rose to 0.5480, 0.1664, and 0.3396 in Table 3

Table 4. Simulated MSE values when $p = 10$ and correlation = 0.99.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	8.9238	2.9746	2.3485	1.9897	0.8160	0.5833
ZINB		8.5397	1.5501	0.6556	1.1539	0.4111	0.1838
HP		9.0270	3.0823	2.5481	2.2561	0.8328	0.6888
RHP		1.6635	0.8744	1.2140	0.5821	0.4467	0.3738
ZIPR		3.0454	0.5430	0.8647	2.1739	1.3109	0.8109
ZIP	1	63.6455	7.3088	4.2165	246.4357	1.1843	1.1361
ZINB		89.9225	3.3579	0.9582	454.4429	0.8452	0.4271
HP		20.3865	9.4403	4.4168	358.9365	1.2579	1.1288
RHP		2.1371	3.5275	1.3503	1.8796	0.4028	0.2683
ZIPR		5.7468	0.0804	0.7059	2.5298	0.7959	0.5566

Table 5. Simulated MSE values when $p = 20$ and correlation = 0.80.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	1.6184	0.3598	0.1940	0.3216	0.1026	0.0771
ZINB		2.4250	0.1441	0.0756	0.3216	0.0790	0.0511
HP		2.2216	0.3335	0.1879	0.3013	0.0982	0.0758
RHP		0.2934	0.1891	0.1043	0.1759	0.0621	0.0482
ZIPR		0.3627	0.3192	0.2239	0.7328	0.2667	0.1987
ZIP	1	35.2743	2.5484	0.3701	44.6856	0.3511	0.0803
ZINB		57.8830	2.5484	0.1473	80.2800	0.3511	0.0662
HP		5.3891	2.1501	0.3006	1.0130	0.2847	0.0802
RHP		1.0451	0.3618	0.1673	0.7842	0.1119	0.0524
ZIPR		10.7141	0.5357	0.3188	2.7517	0.4419	0.3533

Table 6. Simulated MSE values when $p = 20$ and correlation = 0.90.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	16.3951	0.9199	0.4443	0.8697	0.2113	0.1190
ZINB		29.0201	0.4175	0.0936	0.8666	0.1336	0.0565
HP		12.6617	0.8701	0.4032	0.6131	0.2001	0.1297
RHP		0.8312	0.4493	0.2164	0.2431	0.1218	0.0770
ZIPR		1.9536	0.9269	0.6008	1.3550	0.7184	0.5854
ZIP	1	76.0361	2.0485	0.6756	79.2102	0.5676	0.1661
ZINB		57.6063	2.0486	0.2749	74.6771	0.5691	0.0782
HP		43.5074	1.4951	0.6509	56.8193	0.5198	0.1612
RHP		8.0564	0.7636	0.3412	6.7205	0.1917	0.0862
ZIPR		13.6066	3.3632	0.7471	13.3181	1.6514	0.7941

(correlation = 0.95), and further to 1.6635, 0.8744, and 1.2140 in Table 4 (correlation = 0.99), but remained well below those of other models. Similar relative patterns were observed in Tables 5–8 (where $p = 20$). Even at high correlation (0.99), RHP achieved better MSEs than all other models across sample sizes, showcasing its ability to mitigate multicollinearity *via* ridge regularization.

3.2.4. Effect of intercept logit on performance

Scenarios with intercept logit = 1 (as opposed to 0) generally posed more difficulty for standard models, leading to inflated MSEs. However, RHP again exhibited remarkable stability. In Table 1

Table 7. Simulated MSE values when $p = 20$ and correlation = 0.95.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	66.1045	1.5332	0.8185	94.8555	0.1979	0.2296
ZINB		54.3761	0.5588	0.1707	63.6331	0.1494	0.0717
HP		23.0344	1.4863	0.7871	1.1961	0.1991	0.2347
RHP		1.6375	0.7530	0.4048	0.4626	0.1227	0.1269
ZIPR		2.9162	1.9055	1.3847	3.2682	1.6489	1.2257
ZIP	1	89.9291	21.6549	1.5849	104.6026	16.9788	0.2669
ZINB		72.5484	18.1405	0.4337	96.4047	13.0056	0.1234
HP		31.5129	14.8074	1.5402	27.2371	0.8994	0.2653
RHP		10.7392	1.7588	0.7879	11.8241	0.4022	0.1449
ZIPR		17.5275	2.8361	2.1395	20.5567	2.8592	2.2320

Table 8. Simulated MSE values when $p = 20$ and correlation = 0.99.

		Overdispersion parameter = 1			Overdispersion parameter = 5		
		Sample size			Sample size		
Models	Intercept logit	100	200	500	100	200	500
ZIP	0	104.6559	11.3466	3.5794	168.4235	1.2242	0.6409
ZINB		96.5951	2.0988	0.4683	168.6937	0.6403	0.1845
HP		97.6672	11.8289	3.6592	4.3634	1.2212	0.6524
RHP		7.3989	6.0763	1.8352	1.9304	0.7122	0.3670
ZIPR		20.6696	14.7982	5.9604	26.0243	11.1281	5.8779
ZIP	1	166.9572	28.8508	6.7499	291.5065	4.9941	1.1172
ZINB		179.0144	72.3412	1.5498	384.2865	4.9941	0.4427
HP		113.4953	17.9064	6.8062	169.7592	4.3653	1.1478
RHP		18.4937	8.9163	3.4090	24.5913	2.1765	0.5794
ZIPR		32.6322	13.3302	7.4189	49.6148	15.3453	11.3637

(correlation = 0.90, $p = 10$, overdispersion = 1), ZIP and ZINB showed dramatically high MSEs at $n = 100$: 52.6066 and 68.9659, respectively, whereas RHP maintained a low MSE of 0.0189. This trend continued across all levels of correlation and predictor counts. Notably, in Table 8 (correlation = 0.99, $p = 20$), RHP's MSEs under intercept logit = 1 were 18.4937, 8.9163, and 3.4090 for $n = 100$, 200, and 500, compared to ZIP's MSEs of 166.9572, 28.8508, and 6.7499, respectively, highlighting the proposed model's superior handling of intercept shift scenarios.

3.2.5. Effect of overdispersion on performance

Overdispersion substantially impacted models not designed to account for it, such as ZIP. ZINB handled it better than ZIP, but RHP consistently outperformed both. In Table 2 (overdispersion = 1), RHP had an MSE of 0.0729 at $n = 100$, while ZIP and ZINB showed 0.5547 and 0.4486, respectively. When overdispersion increased to 5, the MSEs of ZIP and ZINB spiked considerably, while RHP's MSEs remained stable (0.0667 $n = 100$, Table 2). The model's robustness to overdispersion was even more evident in high numbers of predictors, high correlation, and high overdispersion scenarios (Table 8), where RHP consistently achieved the lowest MSEs, such as 1.9304 (overdispersion = 5, $n = 100$) compared to ZIP (168.4235) and ZINB (168.6937).

4. Real-data analysis

To validate the robustness and practical applicability of the proposed count data models under various complex scenarios, two real-world datasets were analyzed. The analysis aimed to determine if the performance trends from the simulation study apply in real-world settings with multicollinearity, overdispersion, and excess zeros. The real data findings validate and strengthen the simulation results, enhancing the reliability of the comparison among the modeling methods.

4.1. Plastic Plywood data

The Plastic Plywood dataset is employed to evaluate the performance of the models under consideration. This dataset has also been utilized in prior studies by various researchers (Marcondes Filho and Sant'Anna 2016; Sami et al. 2022; Ashraf et al. 2025). It comprises 100 observations with one response variable and four independent variables: volumetric shrinkage, assembly time, wood density, and drying temperature, denoted as x_1, x_2, x_3, x_4, respectively. The response variable measures the number of defects found in each area of laminated Plastic Plywood.

The estimated dispersion parameter is $\hat{\phi} = 2.416$, indicating that the data exhibit overdispersion. This is further supported by the index of dispersion $D = \sigma_y^2/\mu_y$, which was calculated to be 135.614, significantly greater than one, further confirming the presence of overdispersion in the response variable.

To assess multicollinearity, the condition index (Cl) was computed using the formula $\mathrm{CI} = \sqrt{\lambda_{\max}/\lambda_{\min}}$, where $\lambda_{\max}$ and $\lambda_{\min}$ denote the maximum and minimum eigenvalues of the correlation matrix S (excluding the intercept). The resulting Cl value of 8634.73 indicates severe multicollinearity among the explanatory variables. In addition, the correlation matrix, presented in Figure 1, further illustrates high pairwise correlations among the regressors, corroborating the presence of substantial multicollinearity within the dataset.

The analysis of the Plastic Plywood dataset in Table 9 provides strong empirical support for the conclusions drawn from prior simulation studies, confirming the effectiveness of regularized count models in handling zero inflation and overdispersion. Among the models evaluated, the

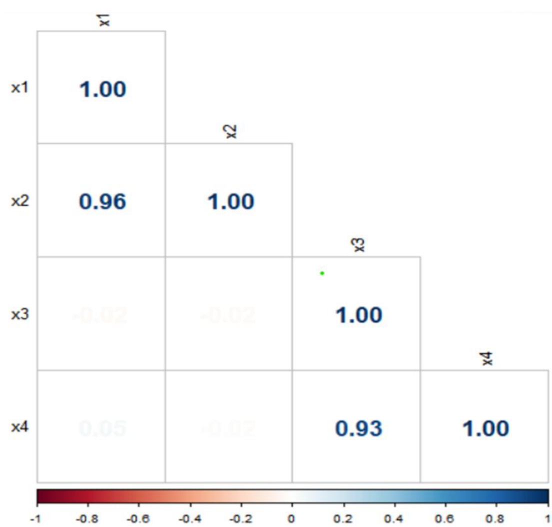

Figure 1. Correlation matrix.

Table 9. MSE values for Plastic Plywood data.

Method	MSE
ZIP	0.5730
ZINB	0.5730
HP	0.2284
ZIPR	0.1007
RHP	0.0901

RHP model demonstrated the best predictive performance, yielding the lowest MSE of 0.0901, followed by the ZIPR model (0.1007). These results represent a substantial improvement over traditional models such as the HP (MSE = 0.2284), ZIP, and ZINB models (both with MSE = 0.5730). The superior performance of ridge-augmented models underscores the value of regularization in stabilizing parameter estimates, mitigating multicollinearity, and enhancing generalization. Thus, the findings from the Plastic Plywood data analysis provide compelling real-world validation of the simulation-based evidence, highlighting the practical advantages of incorporating penalized methods in modeling count data with complex distributional characteristics.

4.2. bioChemist data

The bioChemists dataset from R programming contains information about a sample of 915 graduate students enrolled in biochemistry Ph.D. programs. The response variable, "art," indicates the number of academic articles produced by each student during the last three years of their Ph.D. studies. Key explanatory variables include fem (a categorical variable indicating gender, either "Men" or "Women"), *mar* (marital status: "Single" or "Married"), and *kid5* (a numeric variable counting the number of children aged 5 or younger). Additional predictors related to academic background and environment include *phd*, which reflects the prestige of the student's Ph.D. department, and *ment*, which indicates the number of publications authored by the student's Ph.D. mentor in the same 3-year period (Long 1990).

Initial exploration of the response variable art reveals a significant departure from the assumptions of a standard Poisson model. Specifically, there are 275 observed zeros in article counts, compared to only 191 zeros expected under the Poisson distribution, highlighting strong evidence of zero inflation.

Furthermore, the data exhibit moderate overdispersion, with an estimated dispersion parameter of 2.19, suggesting that the variance in article counts exceeds the mean. This may warrant the use of a negative binomial or zero-inflated model for proper estimation. The condition number of the design matrix is 49.87, indicating moderate multicollinearity among predictors, which should be considered when interpreting model coefficients.

Based on Table 10, among the models assessed, the RHP model yielded the lowest MSE (0.04999), outperforming traditional approaches such as the ZIPR model (0.11589), ZINB model (0.08542), and HP model (0.05253). The incorporation of ridge regularization appears to have improved model stability and predictive power by mitigating multicollinearity and controlling for overfitting. Importantly, the analysis of this real-world dataset validates the findings from earlier simulation studies, reinforcing the conclusion that the RHP model performs more effectively in the presence of zero inflation and moderate overdispersion.

5. Conclusions

In this study, we proposed a novel RHP model that effectively combines the strengths of hurdle count models with the regularization capability of ridge regression. This hybrid framework is specifically designed to address key challenges that frequently arise in count data modeling: zero inflation, overdispersion, and particularly, multicollinearity among predictors. While previous

Table 10. MSE values for *bioChemist* data.

Method	MSE
ZIP	0.11589
ZINB	0.08542
HP	0.05253
ZIPR	0.06419
RHP	0.04999

studies have explored penalized Poisson or negative binomial models (Wang, Wang, et al. 2020), the integration of ridge regularization into the HP structure represents a unique methodological contribution. To the best of our knowledge, this is the first attempt to construct a regularized hurdle-based framework tailored to handle zero-inflated and overdispersed count outcomes under multicollinearity.

Theoretical development of the RHP model was provided, highlighting its flexibility within the generalized linear modeling framework. By incorporating an L_2 penalty term into the HP likelihood, the model not only reduces estimation variance in high-dimensional settings but also enhances prediction stability in the presence of highly correlated covariates. This regularization strategy serves as a safeguard against overfitting and improves model generalizability properties that are essential in applied fields such as epidemiology, transportation, health sciences, and marketing analytics, where complex count data structures are common.

Through simulation experiments and empirical applications, the RHP model demonstrated clear advantages over traditional unpenalized methods. These evaluations provide strong empirical support for its adoption in real-world count data analysis. However, the broader impact of this work lies in its methodological innovation and extensibility. By bridging hurdle modeling and ridge penalization, this study opens a new line of research for developing regularized mixture models that handle structural zeros, overdispersion, and predictor dependencies in an integrated fashion.

Looking forward, several promising directions emerge. First, there is potential to develop an open-source R package that implements the RHP model along with companion tools for parameter tuning, diagnostics, and visualization. Such software would facilitate accessibility and encourage broader application across disciplines. Second, the modeling framework could be extended to robust ridge estimators to handle both multicollinearity and outliers in the Hurdle model, Bayesian settings, nonlinear covariate effects, or machine learning pipelines involving ensemble and deep learning methods for complex count-based prediction tasks (Alghamdi et al. 2025; Kibria et al. 2025; Mohammad et al. 2025). The RHP model could be adapted to longitudinal and spatially correlated count data, where the challenge of excess zeros and multicollinearity persists. Last but not least, a comprehensive parameter-estimation comparison, addressing sign changes, shrinkage patterns, and stability diagnostics, might be considered in future research.

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Author contributions

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ORCID

HM Nayem http://orcid.org/0000-0002-6431-0636
B. M. Golam Kibria http://orcid.org/0000-0002-6073-1978

Data availability statement

No new data were created or analyzed in this study. Data sharing is not applicable to this article.

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